

BERCLAY MANANI AYORA

berclaym@gmail.com

P.O. Box 12215-00100, Nairobi

+254718808824

Objectives

Creative, analytical and motivated B.Sc. Electrical and Electronics Engineering graduate, seeking to grow in hands-on electrical and electronics engineering practical skills, software programming and project conception and management.

Experience

NOVA TRIPZOORI | Mombasa

January 2025 – May 2026 Full-Stack Developer – OTA Tech Dept.

Platform: <https://www.tripzoori.com/>

- Develop and maintain **backend services using Python FastAPI** for a production **OTA booking platform**.
- Design and implement **RESTful APIs** integrating **PostgreSQL databases using SQLAlchemy ORM**.
- Build responsive frontend interfaces using **ReactJS and Tailwind CSS**.
- Implement **end-to-end full-stack features**, integrating frontend components with backend APIs.
- Optimize **database queries and API performance** to improve application efficiency.
- Deploy and manage full-stack applications to **cloud platforms (Fly.io and Netlify)**, handling environment configuration and production releases.
- Collaborate with developers and stakeholders to **deliver new platform features and system improvements**.
- Development and maintenance of the blog WordPress site:
<https://www.blog.tripzoori.com>

SAFARICOM PLC | Nairobi

May 2024 – November 2024 Intern (Internal Audit) – Automation Dept.

- Development of automation software using Flask Python framework and SQL (MySQL).
- Development of productivity applications using Microsoft PowerApps in collaboration with Power Automate and SharePoint lists

- Programming continuous audit and continuous monitoring tests using SQL (Oracle) and Python, and using Excel in test results analysis.
- Data Analysis and visualization using Qlik Sense

EMS CONSULTANTS | Nairobi

April 2024 – May 2024 **Intern (Energy Auditor) – EEE Department**

- Energy data collection from industries including factories.
- Analysis of energy data collected and report compilation.

GOLDMARX SOLUTIONS | Eldoret

March 2023 – July 2023 **Intern**

- Installed and configured **network equipment, CCTV systems, and internet routers** for homes and institutions.
- Performed **hardware troubleshooting and maintenance** for computers, printers, and networking devices.

EATON POWERWARE | P.O. Box 448-00606, Nairobi

February 2022 – April 2022 **Student Intern**

- Assisted in **maintenance and installation of electrical and electronic systems** in both field and workshop environments.
- Designed a **power distribution unit prototype** using cable trays and recycled CPU power inputs.

KENAFRIC INDUSTRIES | P.O. BOX 39257-00623, Nairobi

February 2021 – April 2021 **Student Intern**

- Performed **maintenance of industrial equipment** including motors, heaters, and thermocouples.
- Gained exposure to **Siemens PLC programming and industrial electrical wiring**.

Skills

TECHNICAL

- **Programming Languages**
Python, JavaScript, C++, C
- **Backend Development**
FastAPI, Flask, Django, REST API development

- **Frontend Development**
ReactJS, Tailwind CSS, HTML5, CSS3, Bootstrap
- **Databases**
PostgreSQL, MySQL, MongoDB, Oracle SQL
- **Tools & Platforms**
Git, PowerApps, Power Automate, Qlik Sense, WordPress
- **Engineering & Development Tools**
MATLAB, KiCAD, Proteus, TIA Portal

ADDITIONAL SKILLS

- **Continuous Learning**
Actively expanding knowledge in **Data Science, Machine Learning and AI** through platforms such as Udemy and W3Schools.
- **Adaptability & Collaboration**
Experience working in diverse technical environments including **software development, automation, electrical systems and data analysis.**

Education

UDEMY

May 2026 – Present **CompTIA Security+ (SY0-701)**

- Preparing for the CompTIA Security+ certification exam; studying cybersecurity principles, network defence, risk management, and incident response.

June 2026 – July 2026 **Software Architecture & Design of Modern Large-Scale Systems**

- Studying software architecture principles, system design, scalability, distributed systems, microservices, cloud-native architectures, performance optimization, reliability, and design trade-offs for large-scale applications.

December 2022 – December 2022 **Introduction to Machine Learning**

- Completed an introductory course on Machine Learning

SKILLSOFT – SAFARICOM PLC & CODECADEMY

July 2024 – August 2024 **Data Analytics Foundations (Track 1 & 2)**

- Exploratory data analysis (EDA) with Python
- Python for Data Science
- Data Visualization with Python

July 2024 – August 2024 **Data Analytics for Associates (101)**

- Basics of Data Analytics

June 2024 – July 2024 **Artificial Intelligence for Associates (101)**

- Basics of AI, ethics and advantages in the workplace

May 2024 – June 2024 **Agile for Associates (101)**

- Incorporating the Agile method of software development in the workplace

W3SCHOOLS

February 2023 – March 2023 **MERN Stack, Full-stack**

- Fully developed two sites using MERN; a front-end site using React and another full-stack site with forum (social platform) structure.

KENYATTA UNIVERSITY

September 2017 – July 2023 **B.Sc. Electrical and Electronics Engineering**

Second Class Honors (Lower Division)

- Core areas: **electrical systems, electronics, wireless communication, and embedded systems.**
- Technical exposure to **C/C++ programming, MATLAB, PCB design (KiCAD, Proteus).**
- Member of **IEEE student community**, participating in technical workshops and innovation programs.

ZALEGO INSTITUTE OF TECHNOLOGY AND INNOVATION

March 2017 - July 2017 **Certificate – Windows, Android, Web Design using HTML, CSS and PHP**

- Building websites from scratch based on assignments and lectures, using PHP, JavaScript, HTML, CSS and Bootstrap.
- More projects done in other languages can be found on my GitHub page, <https://github.com/barcs-manani>

Certificates

All my certificates can be accessed from Google Drive using the links below for verification:

- Agile, Safaricom -
<https://drive.google.com/file/d/17mWjfW9KzCcjm2wmnIOBXoYiTY6Vk7Qq/view?usp=sharing>
- Data Analytics for Associates, Safaricom -
https://drive.google.com/file/d/19HbjzgU_fGtPCeMevSqlqeV4dC1SFSIx/view?usp=sharing
- Degree Certificate, Kenyatta University -
<https://drive.google.com/file/d/1GMV8OLHLi6qjqaPdue3j3R2eBfI2EtUW/view?usp=sharing>
- KCSE Certificate, Moi High School Kabarak -
<https://drive.google.com/file/d/1ABZEKEWTq0ck82wMiveagydrySq80tMS/view?usp=sharing>
- Software Architecture & Design of Modern Large-scale Systems -
https://drive.google.com/file/d/1EBuivOujtFsWbZzNUTAQgXltYv_XcFuZ/view?usp=sharing